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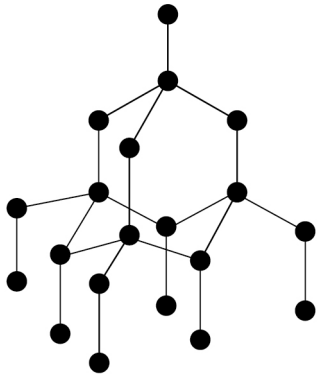
0 5 This question is about structure and bonding.

0 5 . 1 **Figure 5** shows part of the structure and bonding in diamond.

Figure 5



Explain why diamond has a high melting point.

[3 marks]

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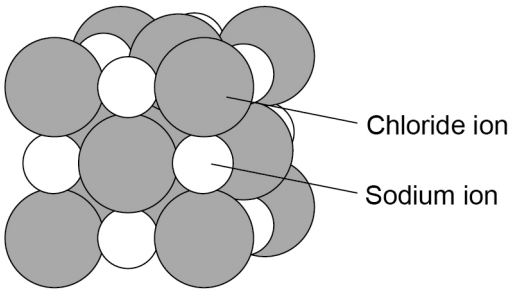
0 5 . 2 **Figure 6** shows part of the structure and bonding in sodium chloride (NaCl).

Figure 6



Explain the conditions needed for sodium chloride to conduct electricity.

[3 marks]

Question 5 continues on the next page

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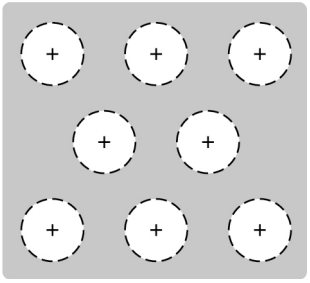
0 5 . 3 **Figure 7** shows the structure of sodium.

Figure 7



Describe how sodium conducts thermal energy.

[3 marks]

9

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0 6 Group 2 metal carbonates thermally decompose to produce a metal oxide and a gas.

0 6 . 1 Give the formula of each product when calcium carbonate ($CaCO_3$) is heated.

[2 marks]

_____ and _____

0 6 . 2 The relative formula mass (M_r) of a Group 2 metal carbonate is 197

Relative atomic masses (A_r): C = 12 O = 16

Calculate the relative atomic mass (A_r) of the Group 2 metal in the metal carbonate.

Name the Group 2 metal.

[3 marks]

_____

_____

_____

_____

_____

_____

Relative atomic mass (A_r) = _____

Metal _____

Question 6 continues on the next page